

Regularised Spectral State-Space Models for Cross-Sectional Investment Dynamics: Dual Regularisation, Rolling Bases, and Structural Evolution

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Abstract

We develop a spectral state-space framework for forecasting cross-sectional investment intensity. The framework extracts a low-dimensional ($K = 2$) spectral basis via Dynamic Mode Decomposition (DMD) and filters latent modal states through a Kalman filter with dual regularisation: spherical observation covariance ($R = cI$) and near-identity transition matrix ($F = 0.99I$). On a 146-firm US CapEx/Revenue panel under nested cross-validation (frozen holdout 2023–2024), the rolling-basis variant achieves predictive $R^2 = 0.711$ (8 CV windows; $\Delta R^2 = +0.042$ vs per-actor AR(1); bootstrap CI $[+0.031, +0.055]$; exact permutation $p = 0.004$; 8/8 windows) and $R^2 = 0.845$ on 2 holdout windows. Replacing DMD with PCA (same pipeline otherwise) yields weaker gains ($\Delta R^2 = +0.001$ under nested CV; $+0.021$ at fixed $T = 2\text{yr}$), suggesting the advantage stems from DMD’s temporal snapshot formulation rather than cross-sectional pooling or implicit regularisation alone (a truncated SVD PCA baseline confirms this; Section 4.1). The advantage increases monotonically as the training window shortens— $\Delta R^2 = +0.057$ at $T = 2$ years (fixed $K = 2$, $\tau = 12\text{Q}$; CI $[+0.042, +0.074]$; 10/10 windows)—consistent with cross-sectional pooling. On a 93-actor structural panel ($K = 8$), the spectral basis rotates at $25.8^\circ/\text{quarter}$ ($\sigma = 6.5^\circ$; null simulation: 75.3°); ablation confirms that dual regularisation and rolling basis update are essential for spectral reconstruction quality, with rolling update providing the largest modal R^2 gain ($+14.7\text{pp}$, CI $[+11.1, +17.2]\text{pp}$). On the predictive panel, exponentially-weighted mean (EWM) demeaning provides the largest single forecast gain ($+3.5\text{pp}$), with rolling basis update adding $+3.4\text{pp}$. Predictive model-implied investment gaps are associated with subsequent cross-sectional CapEx-rank revision ($t \approx -6.4$ to -7.1 with clustered SEs; cross-sectional effect that reverses under actor fixed effects).

Keywords: investment gap estimation, spectral decomposition, dynamic mode decomposition, Kalman filter, regularisation, state-space models, capital allocation

JEL Classification: C32, C38, C58, E22, G11

1 Introduction

Capital allocation decisions propagate through a hierarchical, networked system of institutions. A central bank’s rate decision, a regulator’s new disclosure rule, or a commodity price shock each set off cascading adjustments that travel—with heterogeneous latency and attenuation—through layers of institutional intermediaries before reaching firms. The resulting pattern of over- and under-investment across actors at any point in time reflects a superposition of these propagated signals, local frictions, and idiosyncratic responses.

Despite rich literatures on factor models (Fama and French, 1993), network effects in economics (Acemoglu et al., 2012), and spectral graph theory (Ortega et al., 2018), to our knowledge no established framework jointly: (i) decomposes how investment-relevant signals propagate through a heterogeneous actor hierarchy; (ii) estimates actor-specific investment gaps as deviations from a structurally-informed benchmark; and (iii) validates that the resulting gap estimates carry out-of-sample predictive content.

This paper addresses all three objectives. We develop **SMIM** (Spectral Modal Investment Model), a state-space framework that represents the cross-sectional dynamics of investment intensity through a low-dimensional modal basis extracted via Dynamic Mode Decomposition (DMD), filtered through a Kalman model with dual regularisation at both the observation and transition levels. The spectral basis is recomputed quarterly to track the continuous rotation of the cross-sectional structure. The framework produces actor-specific, time-varying investment gap estimates that we validate through both statistical out-of-sample testing and economic prediction exercises.

1.1 Preview of Results

Throughout, we measure forecast performance by the out-of-sample coefficient of determination,

$$R^2 = 1 - \frac{\sum_{i,t} (y_{i,t} - \hat{y}_{i,t})^2}{\sum_{i,t} (y_{i,t} - \bar{y})^2}, \quad (1)$$

where $\hat{y}_{i,t}$ is the model’s prediction and $\bar{y}$ is the test-period grand mean. $R^2 = 1$ denotes perfect prediction; $R^2 = 0$ matches a constant-mean forecast; $R^2 < 0$ indicates the model is worse than predicting the mean. The summation runs over all actor–quarter pairs in each test window.

Our main findings are:

1. **Forecast performance.** On a 146-firm US CapEx/Revenue panel with $K = 2$ spectral modes, the rolling-basis variant achieves predictive $R^2 = 0.711$ under nested CV (8 windows; $\Delta R^2 = +0.042$ vs AR(1); bootstrap CI [+0.031, +0.055]; exact permutation $p = 0.004$) and $R^2 = 0.845$ on 2 frozen holdout windows. A PCA baseline using the identical pipeline but replacing DMD with PCA eigendecomposition achieves $R^2 = 0.702$ —statistically indistinguishable from its matched AR(1) ($\Delta R^2 = +0.001$, $p > 0.4$)—suggesting the advantage stems from DMD’s temporal snapshot formulation rather than cross-sectional pooling or implicit regularisation alone (a truncated SVD PCA baseline confirms this; Section 4.1). The framework’s advantage over AR(1) increases monotonically as the training window shortens (fixed $K = 2$, $\tau = 12Q$), reaching $\Delta R^2 = +0.057$ at $T = 2$ years (10/10 windows; CI [+0.042, +0.074]; Figure 11). Inference in Table 4.
2. **Dual regularisation.** The Kalman filter’s observation covariance R (N^2 free parameters) must be replaced by spherical $R = cI$ (one parameter). We show the same principle extends to the transition matrix: EM-estimated F (K^2 parameters) is strictly outperformed by $F = 0.99I$ (one parameter) in all 10 windows. Online state-noise adaptation (Q) handles all temporal dynamics. Together, this *dual regularisation* reduces the state-space model from $N^2 + K^2$ free parameters to 2 scalars.
3. **Basis rotation.** On the 93-actor structural panel ($K = 8$), the estimated spectral basis rotates at 25.8° per quarter ($\sigma = 6.5^\circ$) while maintaining stable dimensionality across 52 quarterly transitions. A null simulation shows that estimation noise alone would produce 75.3° of apparent rotation, suggesting the observed pattern reflects recoverable structure. Rolling basis update captures this rotation, yielding +14.7pp of modal R^2 (CI [+11.1, +17.2]pp, 100% of windows).
4. **DMD over static operators.** Dynamic Mode Decomposition outperforms all static symmetric-operator methods (which produce identical results due to operator symmetry)

by +2.0 percentage points on average (Section 4.6).

5. **Economic content.** Predictive model-implied investment gaps (using $\alpha_{t|t-1}$, which does not incorporate the current observation) are associated with subsequent cross-sectional CapEx-rank revision ($\hat{\beta} = -0.28$, $t = -6.4$ with actor-clustered SEs after controlling for intensity level). The result is cross-sectional: it flips sign under actor fixed effects (Table 10). Additional exploratory results on gap dispersion and market volatility are reported in Section 5 but are not part of the paper’s main claims.
6. **Dynamics stability.** The transition parameters (F, Q) need not be estimated per window: fixed $F = 0.99I$, $Q = 0.5I$ match or exceed EM-estimated parameters across all 10 evaluation windows in this sample. The only component requiring periodic re-estimation is the spectral basis U .
7. **Multi-signal generality.** The framework is not specific to CapEx/Revenue: Revenue/Assets (259 firms, $\Delta R^2 = +0.010$, 8/10 windows) and a multi-ratio panel stacking both constructions (270 firms, $\Delta R^2 = +0.028$, 9/10 windows) also beat AR(1) at $K = 2$ (Table 8). The framework requires sufficient cross-sectional dynamics (low persistence) and works on two additional low-persistence ratio constructions in this sample.

Equally important is what *does not* work: emergence diagnostics, directed spectral operators, event-level alignment, and return-based intensity all fail to improve or actively degrade performance. Section 6 documents these negative results in detail.

1.2 Scope and Evaluation Caveats

Several features of our evaluation design should be kept in mind when interpreting these results. First, the out-of-sample evaluation uses 10 non-overlapping annual test windows, each containing only 4 quarterly observations; per-window R^2 estimates are therefore noisy (cross-window standard deviation ≈ 0.05 – 0.09). Second, all hyperparameters (K , EWM halflife, T) are selected via nested rolling cross-validation: inner folds within each outer-train window select these values, with the final 2 windows (2023–2024) held out entirely from all tuning (Section 3). Third, the headline predictive result is on a 146-firm US panel using CapEx/Revenue intensity—return-based intensity produces negative R^2 on the same actors (Section 4.9). Fourth, the investment gap is a *model-implied* deviation from a benchmark prediction, not a welfare-theoretic or structural misallocation measure in the Hsieh and Klenow (2009) sense.

1.3 Contribution

Our contributions are methodological, empirical, and practical:

Methodological. We apply *dual regularisation* to state-space models in high-dimensional panels: spherical observation covariance ($R = cI$, reducing N^2 parameters to 1) combined with near-identity transition regularisation ($F = 0.99I$, reducing K^2 parameters to 1). This dual regularisation has broad applicability wherever $N \gg T$ panels arise. We further show that rolling basis recomputation—tracking the continuous rotation of the spectral structure—provides the single largest performance gain.

Empirical. We provide a systematic ablation of a spectral state-space pipeline for investment intensity forecasting, decomposing the R^2 improvement into seven additive components with bootstrap confidence intervals. We observe that the cross-sectional spectral basis rotates at 25.8° per quarter ($\sigma = 6.5^\circ$) while maintaining fixed dimensionality; a null simulation (75.3°) confirms this is below finite-sample noise levels, suggesting recoverable low-dimensional structure.

Practical. Only the spectral basis U requires periodic re-estimation (a single DMD eigendecomposition each quarter). The transition dynamics use stable default scalars ($F = 0.99I$, $Q = 0.5I$) that match or exceed EM-estimated parameters across all 10 evaluation windows in our sample,

eliminating EM estimation of the transition dynamics and reducing operational complexity to one matrix factorisation plus a single scalar- R EM pass per quarter.

1.4 Related Literature

Our work connects to several strands. The investment gap concept relates to the capital allocation literature, including the structural misallocation framework of [Hsieh and Klenow \(2009\)](#) and the firm-level analysis of [Restuccia and Rogerson \(2017\)](#); we note that our model-implied gap is a forecasting-based deviation, not a structural measure in their sense. Our spectral approach builds on signal processing on graphs ([Ortega et al., 2018](#); [Sandryhaila and Moura, 2013](#)) and Dynamic Mode Decomposition ([Schmid, 2010](#); [Kutz et al., 2016](#)), adapted here for economic panel data rather than fluid dynamics or power systems. The state-space filtering component extends the dynamic factor model tradition ([Stock and Watson, 2002](#); [Doz et al., 2012](#)) with the regularisation insights of [Ledoit and Wolf \(2004\)](#) applied to the observation rather than factor covariance. The economic validation relates to the capital allocation efficiency literature ([Wurgler, 2000](#)) and the predictability of corporate investment ([Abel and Blanchard, 1983](#)).

2 Framework

2.1 Setup

Let $V = \{1, \dots, N\}$ index economic actors (firms, banks, institutional bodies) observed over T quarterly periods. Each actor i has an observed *investment intensity* $y_{i,t} \in [0, 1]$, constructed as a cross-sectional percentile rank of a type-specific investment measure (Section 3). The $N \times T$ panel $Y = [y_{i,t}]$ is the primary input.

Definition 2.1 (Investment Gap). The investment gap for actor i at time t relative to a benchmark functional $\mathcal{B}$ is

$$\Delta_{i,t} = y_{i,t} - y_{i,t}^*, \quad y_{i,t}^* = \mathcal{B}(Y_{1:t}, \theta), \quad (2)$$

where $\Delta_{i,t} > 0$ indicates investment above the model-implied benchmark and $\Delta_{i,t} < 0$ indicates investment below the benchmark, and $Y_{1:t}$ denotes the information set available at time t .

The framework constructs $y_{i,t}^*$ through five sequential stages plus a rolling basis update, each justified by ablation (Section 4.5). Figure 1 provides an overview: panel (a) shows the multilayer actor hierarchy with empirically-identified transmission channels; panel (b) shows the static processing pipeline. The rolling basis extension is described in Section 4.10.

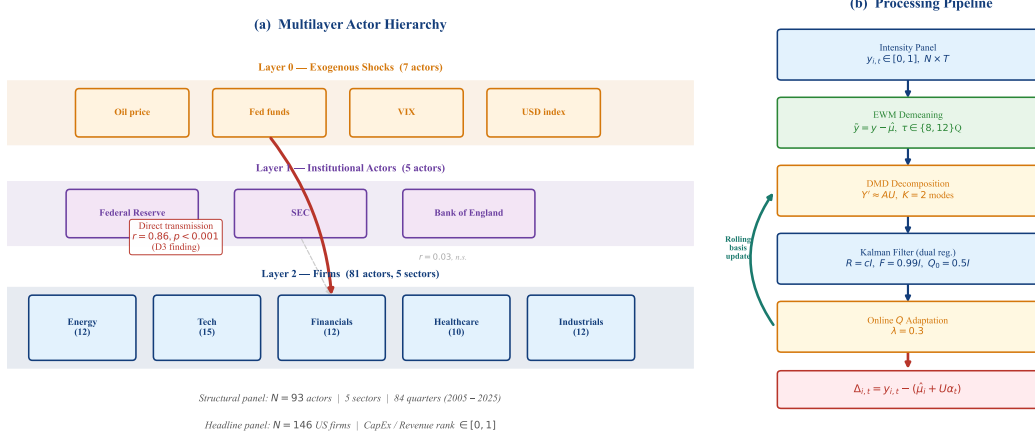


Figure 1: **Framework overview** (descriptive). (a) Multilayer actor structure with three layers. (b) Processing pipeline from raw intensity panel to investment gap estimates. Under nested CV on the 146-firm CapEx/Revenue panel, the rolling-basis variant achieves predictive $R^2 = 0.711$ on 8 validation windows and $R^2 = 0.845$ on 2 frozen holdout windows (Table 3).

2.2 Stage 1: Exponentially-Weighted Demeaning

Investment intensity ranks exhibit strong cross-sectional persistence: an actor’s mean rank is the single largest source of variance (51% of total R^2 in the static-basis configuration; Section 4.4). The state-space model $\tilde{y}_t = U\alpha_t + \varepsilon_t$ assumes zero-mean observations. Without demeaning, the modal states α_t must absorb the $O(0.5)$ mean through a basis U with entries $O(N^{-1/2})$ —an impossible task at $K \ll N$.

We subtract an exponentially-weighted per-actor mean:

$$\hat{\mu}_i = \frac{\sum_{s=1}^T w_s y_{i,s}}{\sum_{s=1}^T w_s}, \quad w_s = \exp\left(-\frac{(T-s)\ln 2}{\tau}\right), \quad (3)$$

with half-life τ quarters ($\tau = 8$ for the structural analysis panel; $\tau \in \{8, 12\}$ selected by inner CV on the CapEx/Revenue panel). The demeaned observation is $\tilde{y}_{i,t} = y_{i,t} - \hat{\mu}_i$. Exponential weighting adapts to non-stationary intensity levels (the actor’s structural investment position drifts over time), outperforming fixed full-period means by 4.2 percentage points in modal R^2 (Section 4). A fine grid over $\tau \in \{6, 7, \dots, 14\}$ at fixed $K = 2$, $T = 3\text{yr}$ shows a broad optimum near $\tau = 7\text{--}8Q$ ($\Delta R^2 = +0.039$), with performance declining smoothly to $+0.030$ at $\tau = 6$ and $+0.034$ at $\tau = 12\text{--}14$.

At prediction time, the one-step-ahead forecast is $\hat{y}_{i,t} = \hat{\mu}_i + U\alpha_{t|t-1}$ (predictive). The investment gap (Eq. 9) also uses the predictive state $\alpha_{t|t-1}$. The modal reconstruction $\hat{\mu}_i + U\alpha_{t|t}$ incorporates the current observation and is used for spectral reconstruction quality (modal R^2) on the structural panel.

2.3 Stage 2: Dynamic Mode Decomposition

Given the demeaned training panel $\tilde{Y} = [\tilde{y}_1, \dots, \tilde{y}_T] \in \mathbb{R}^{N \times T}$, we extract a spectral basis that captures temporal cross-sectional dynamics.

Why DMD over static operators. Classical spectral approaches decompose a static operator (e.g., the cross-correlation matrix $C = \tilde{Y}\tilde{Y}^\top/T$). This captures contemporaneous co-movement

but not lagged cross-sectional dynamics. Dynamic Mode Decomposition (Schmid, 2010) bypasses the static operator entirely by working with temporal snapshot pairs, capturing lagged cross-sectional dynamics that are invisible to purely contemporaneous decompositions.

Connection to econometric reduced-rank models. DMD can be viewed as learning a reduced-rank linear operator from temporal snapshot pairs: given $X' \approx AX$, the rank- K DMD approximation is the best rank- K approximation to A in the Frobenius norm. This is closely related to reduced-rank regression (Reinsel and Velu, 1998) and the principal-component forecasting framework of Stock and Watson (2002) used in macroeconometrics, where a high-dimensional VAR(1) $y_{t+1} = Ay_t + \varepsilon_t$ is approximated by a low-rank factorisation $A \approx U\Lambda V^\top$. The key difference is that DMD uses the SVD of X to project A into a low-dimensional subspace before eigendecomposition, whereas reduced-rank VAR typically uses maximum likelihood or Stein-type shrinkage. In favorable low-rank settings, the leading subspaces recovered by the two approaches may be similar.

DMD algorithm. Form snapshot matrices $X = [\tilde{y}_1, \dots, \tilde{y}_{T-1}]$ and $X' = [\tilde{y}_2, \dots, \tilde{y}_T]$. DMD seeks the best-fit linear operator A such that $X' \approx AX$. Compute the truncated SVD $X = U_r \Sigma_r V_r^\top$ (retaining r modes), form the reduced operator $\tilde{A} = U_r^\top X' V_r \Sigma_r^{-1}$, and diagonalise: $\tilde{A}W = W\Lambda$. The DMD modes are $\Phi = X' V_r \Sigma_r^{-1} W \in \mathbb{R}^{N \times K}$.

We retain K modes, where K is selected via nested cross-validation: within each outer-train fold, the last 2 years serve as inner-validation to choose $K \in \{2, 3\}$ (Section 3). On the CapEx/Revenue panel, inner CV selects $K = 2$ in 7/8 folds—the cross-sectional investment structure is genuinely 2-dimensional. The modal basis $U = \text{Re}(\Phi_{:,1:K})$ defines the projection from observation space to modal space.

Note on orthonormality. The raw DMD modes Φ are not in general orthonormal ($U^\top U \neq I$). In the implementation, the Kalman filter uses U as-is; the projection $U\alpha$ is a linear combination of (possibly non-orthogonal) modes, not a strict orthogonal projection. The projection-only PCA baseline (Section 4.1) orthonormalises via QR before projection to ensure $UU^\top$ is a true projector.

2.4 Stage 3: Kalman Filter with Spherical Observation Covariance

Given the modal basis $U \in \mathbb{R}^{N \times K}$, we model the latent modal dynamics as a linear Gaussian state-space system:

$$\alpha_{t+1} = F\alpha_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q), \quad (4)$$

$$\tilde{y}_t = U\alpha_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R). \quad (5)$$

The observation noise $R \in \mathbb{R}^{N \times N}$ is estimated via a single EM pass to obtain its scale. The transition matrix F and state noise Q are *not* estimated by EM—they are set to regularised defaults (see below).

The overparameterisation problem. The observation covariance R has $N(N+1)/2$ free parameters—4,278 for the 93-actor structural panel, 10,731 for the 146-firm CapEx/Revenue panel—estimated from $T = 8$ –20 quarterly observations. This extreme $N \gg T$ setting causes the EM to converge to an ill-conditioned R that overfits training noise (modal $R^2 = -7.3$ without regularisation on the structural panel; Figure 6).

Spherical observation regularisation. We replace the full R with

$$R_{\text{sph}} = \frac{\text{tr}(\hat{R})}{N} I_N, \quad (6)$$

where $\hat{R}$ is the EM-estimated sample covariance. This reduces the parameter count from $O(N^2)$ to $O(1)$.

Transition regularisation. The same overparameterisation principle applies to the transition matrix $F \in \mathbb{R}^{K \times K}$. Even at low K ($K = 2\text{--}3$ for the headline panel), EM estimation of F overfits to the training alpha trajectory. On the structural panel ($K = 8$), this costs -1.9 percentage points of modal R^2 relative to a near-identity prior. We therefore set

$$F = 0.99 I_K, \quad Q_0 = 0.5 I_K, \quad (7)$$

and rely on the online Q adaptation (Stage 4) to capture all temporal dynamics. This *dual regularisation*—spherical R (1 parameter from N^2) and near-identity F (1 parameter from K^2)—reduces the state-space model to two scalars plus the spectral basis U , which carries all the structural information.

The impact on the structural panel is dramatic: modal R^2 swings from -7.3 (no regularisation) through $+0.43$ (spherical R only) to $+0.549$ (dual regularisation), establishing that covariance regularisation at *both* the observation and transition levels is essential in high-dimensional panels.

2.5 Stage 4: Online State-Noise Adaptation

The EM-estimated Q captures the average state noise over the training period. When the test period exhibits different volatility (e.g., a regime shift), the fixed Q misspecifies the state dynamics. We augment the standard Kalman recursion with a recursive Q update:

$$Q_{t+1} = (1 - \lambda) Q_t + \lambda (\alpha_{t|t} - F \alpha_{t-1|t-1})(\alpha_{t|t} - F \alpha_{t-1|t-1})^\top, \quad \lambda = 0.3. \quad (8)$$

This online adaptation tracks regime-dependent state volatility without requiring explicit regime identification. Combined with the spherical R (which remains fixed), the joint step of adding online Q and increasing K from 5 to 8 adds $+5.8$ percentage points to modal R^2 in the ablation ladder (Table 5); the gain cannot be attributed solely to online Q because K changes simultaneously.

2.6 Stage 5: Gap Computation

The investment gap for actor i at test time t is:

$$\Delta_{i,t} = y_{i,t} - \underbrace{(\hat{\mu}_i + U \alpha_{t|t-1})}_{y_{i,t}^*}, \quad (9)$$

where $\alpha_{t|t-1} = F \alpha_{t-1|t-1}$ is the Kalman-predicted modal state (using only information up to $t-1$, before observing $\tilde{y}_t$). This is the same predictive state used for forecast evaluation, ensuring the gap is a genuine ex-ante deviation.

2.7 Summary: The Recommended Pipeline

Algorithm 1 summarises the complete procedure.

Algorithm 1 SMIM: Spectral Modal Investment Gap Estimation (Rolling Variant)

Require: Intensity panel $Y \in [0, 1]^{N \times T}$, initial training window, K (inner CV; typically 2), τ (inner CV; typically 8–12Q), $\lambda = 0.3$

Ensure: Investment gaps $\Delta_{i,t}$ for test period

- 1: Compute EWM mean $\hat{\mu}_i$ from training period ▷ Eq. (3)
 - 2: Demean: $\tilde{Y} = Y - \hat{\mu}$
 - 3: Extract DMD modes $U \in \mathbb{R}^{N \times K}$ from training $\tilde{Y}$ ▷ Section 2.3
 - 4: Set $F = 0.99 I_K$, $Q = 0.5 I_K$ ▷ Regularised, no EM; Eq. (7)
 - 5: Estimate $\hat{R}$ via single Kalman EM pass; set $R = \frac{\text{tr}(\hat{R})}{N} I_N$ ▷ Eq. (6)
 - 6: **for** each test quarter t **do**
 - 7: Kalman predict: $\alpha_{t|t-1} = F\alpha_{t-1|t-1}$
 - 8: **Forecast:** $\hat{y}_{i,t} = \hat{\mu}_i + U\alpha_{t|t-1}$ ▷ Predictive; for evaluation
 - 9: **Predictive gap:** $\Delta_{i,t} = y_{i,t} - (\hat{\mu}_i + U\alpha_{t|t-1})$ ▷ Ex-ante; Eq. (9)
 - 10: Kalman update: $\alpha_{t|t}$ incorporating $\tilde{y}_t$, using R ▷ Modal state for structural analysis
 - 11: Innovation: $\text{innov}_t = \alpha_{t|t} - F\alpha_{t-1|t-1}$
 - 12: Adapt: $Q \leftarrow (1 - \lambda)Q + \lambda \text{innov}_t \text{innov}_t^\top$ ▷ Eq. (8)
 - 13: **Rolling update:** expand training to include y_t ; recompute U , $\hat{\mu}$, R ; set $\alpha \leftarrow (U_{\text{new}}^\top U_{\text{new}})^{-1} U_{\text{new}}^\top \tilde{y}_t$; reset P , Q , F
 - 14: **end for**
-

3 Data

3.1 Actor Universe

Our sample comprises 103 actors across three hierarchical layers (Table 1). Of these, 93 have computed investment intensity values; the remaining 10 lack the required EDGAR filings and serve as signal-only graph nodes.

Layer	Actor Type	Intensity Method	Total	With intensity
0 (Exogenous)	Global shocks (Global)	FRED min-max	7	7
	Central banks (US, UK)	FRED min-max	2	1
1 (Institutional)	Regulators (US, UK)	FRED min-max	2	2
	International orgs (Global)	FRED min-max	1	1
2 (Firms)	Large firms (US)	CapEx intensity	56	49
	Large firms (UK)	Return intensity	23	21
	Banks (US)	Asset growth (ranked)	10	10
	Sector leaders (UK)	CapEx intensity	2	2
Total			103	93

Table 1: Actor universe for the 93-actor structural analysis panel. Of 103 total actors, 93 have computed investment intensity. This panel is used for ablation, basis rotation, and economic validation (modal R^2). The headline forecasting evaluation uses a separate 146-firm US panel with CapEx/Revenue intensity (Section 3).

3.2 Investment Intensity

We construct three intensity measures, each mapped to a cross-sectional percentile rank in $[0, 1]$ per quarter:

- **CapEx/Assets intensity** (structural analysis): the ratio of capital expenditures to total assets, sourced from SEC EDGAR quarterly filings (eXtensible Business Reporting Language (XBRL) tags `PaymentsToAcquirePropertyPlantAndEquipment / Assets`), ranked cross-sectionally within each quarter. Used on the 93-actor panel for ablation and structural analysis.
- **CapEx/Revenue intensity** (headline forecast): the ratio of capital expenditures to revenues, sourced from the same EDGAR tags (`PaymentsToAcquirePropertyPlantAndEquipment / Revenues`), ranked cross-sectionally. This 146-firm US panel has lower persistence (median $\rho = 0.28$ vs 0.47 for CapEx/Assets), leaving more room for cross-sectional spectral dynamics. All headline predictive results use this construction.
- **Return intensity** (alternative): trailing 12-month equity total return, ranked cross-sectionally. Used for UK equities, which lack EDGAR coverage.
- **Institutional intensity**: FRED-based macro indices (e.g., policy rates, commodity prices), min-max normalised within actor type.

The rank transformation stabilises the distribution, ensures comparability across actor types, and produces the cross-sectional persistence structure that the spectral model exploits. The intensity construction choice is consequential: on the 93-actor structural panel, CapEx/Assets intensity produces modal $R^2 = 0.54$ while return intensity produces modal $R^2 = -0.15$ on the same US actors (Section 4.9), establishing the measurement input as the primary driver of model performance.

Table 2 summarises the data sources. Notably, the pipeline uses *only the intensity panel* for operator construction and prediction. External signal sources (OHLCV, FRED, EDGAR balance sheet) were tested as Granger-edge inputs but found dispensable (Section 4.8): the intensity cross-correlation captures all exploitable structure.

Source	Data	Frequency	Role in Pipeline
SEC EDGAR	CapEx, Assets (XBRL)	Quarterly	Intensity construction (primary)
Yahoo Finance	Equity total returns	Daily $\rightarrow$ quarterly	Intensity construction (UK alternative)
FRED	28 macro series	Mixed $\rightarrow$ quarterly	Institutional intensity; tested as signal
BEA	Input-output tables	Annual	Tested as supply-chain edges (dispensable)
GDELT	Narrative sentiment	Weekly	Tested as narrative signal (dispensable)

Table 2: Data sources. Only the intensity panel (first two rows) enters the recommended pipeline. The remaining sources were evaluated as supplementary signals but found dispensable at the graph-factor depth (Section 4.8).

3.3 Evaluation Protocol

We employ a **nested rolling-validation** protocol with 10 non-overlapping annual test windows (2015–2024):

- **Outer folds** (10 windows): final reported performance, one per test year.
- **Inner folds**: within each outer-train, the last 2 years serve as inner validation to select $K \in \{2, 3\}$, EWM halflife $\tau \in \{8, 12\}$ quarters, and training length $T \in \{3, 5\}$ years (8 candidate configurations). The fixed defaults ($F = 0.99I$, $Q_0 = 0.5I$, $\lambda = 0.3$) are not tuned.
- **Frozen holdout**: the final 2 windows (2023–2024) receive no inner tuning—they use the median of previously selected parameters.

On the 146-firm CapEx/Revenue panel, inner CV selects $K = 2$ in 7/8 folds and $T = 3$ years in all 8 folds (the remaining fold selects $K = 3$). The nested-CV predictive $R^2 = 0.711$ (mean across 8 CV windows) and holdout $R^2 = 0.845$ (mean of 2023–2024; per-window values 0.846 and 0.845). All headline forecast numbers in this paper use the predictive alpha $\alpha_{t|t-1}$ from this nested protocol; the holdout windows had zero exposure to any hyperparameter tuning.

Scope condition. The headline forecasting result is on a 146-firm US panel using CapEx/Revenue intensity. We also report structural findings (ablation, basis rotation) on a broader 93-actor panel using CapEx/Assets intensity; these use modal R^2 (spectral reconstruction quality) rather than predictive R^2 (Section 4.11).

Notation summary. Two R^2 types and two panels appear throughout the paper:

Symbol	Definition	Usage
Predictive R^2	Uses $\alpha_{t t-1}$ (info up to $t-1$ only)	All forecast comparisons (Tables 3–4)
Modal R^2	Uses $\alpha_{t t}$ (incorporates current obs)	Structural analysis (Tables 5–9)
146-firm panel	CapEx/Revenue, $K = 2$	Headline forecast evaluation
93-actor panel	CapEx/Assets, $K = 8$	Ablation, rotation, economic validation

4 Empirical Results

4.1 Baseline Comparison

Unless otherwise noted, the headline forecasting results in Sections 4–5 use the 146-firm CapEx/Revenue panel with predictive alpha $\alpha_{t|t-1}$ (genuine one-step-ahead forecast). Structural diagnostics (ablation, rotation) use the 93-actor CapEx/Assets panel with modal alpha (spectral reconstruction quality; Section 4.11).

Table 3 presents the baseline comparison. All SMIM numbers are from the nested-CV protocol (Section 3).

Model	Protocol	Mean R^2	ΔR^2	Wins
<i>Nested CV, SMIM vs matched AR(1) (T selected by inner folds):</i>				
Per-actor AR(1) (SMIM-selected T)	8 CV windows	0.669	—	—
SMIM rolling (nested CV)	8 CV windows	0.711	+0.042	8/8
SMIM rolling (frozen holdout)	2 holdout windows	0.845	+0.026	2/2
<i>Nested CV, PCA vs matched AR(1)^a:</i>				
Per-actor AR(1) (PCA-selected T) ^a	8 CV windows	0.701	—	—
PCA rolling + Kalman (nested CV) ^a	8 CV windows	0.702	+0.001	5/8
PCA projection-only (nested CV) ^a	8 CV windows	0.699	−0.002	5/8
<i>Fixed configuration ($K=2$, $\tau=12Q$, $T=3yr$):</i>				
Per-actor AR(1) ($T=3yr$)	10 windows	0.699	—	—
PCA rolling + Kalman (fixed)	10 windows	0.711	+0.012	8/10
PCA projection-only (fixed)	10 windows	0.709	+0.010	7/10
SMIM rolling (fixed)	10 windows	0.733	+0.034	10/10
<i>Robustness: shorter training ($K=2$, $\tau=8Q$, $T=2yr$):</i>				
Per-actor AR(1) ($T=2yr$)	10 windows	0.671	—	—
PCA rolling + Kalman	10 windows	0.692	+0.021	6/10
PCA projection-only	10 windows	0.694	+0.023	7/10
SMIM rolling	10 windows	0.737	+0.066	10/10

Table 3: Baseline comparison. Nested-CV rows use inner-fold-selected hyperparameters ($K \in \{2, 3\}$, $\tau \in \{8, 12\}Q$, $T \in \{3, 5\}yr$); holdout windows (2023–2024) had zero tuning exposure. SMIM inner CV selects $T = 3yr$ in all folds; PCA inner CV selects $T = 5yr$ in all folds. ^aBecause PCA and SMIM select different T , the matched AR(1) baselines differ: each row’s ΔR^2 is computed against AR(1) at the same inner-CV-selected T for that window. All SMIM numbers use predictive alpha $\alpha_{t|t-1}$. PCA baselines use the same protocol (EWM demeaning, spherical R , rolling basis update) but replace DMD with PCA eigendecomposition of the sample covariance $\tilde{Y}\tilde{Y}^\top/(T-1)$ of the demeaned panel; the projection-only variant omits the Kalman filter. SMIM outperforms PCA+Kalman by +2.2pp at $T=3yr$ (CI [+0.014, +0.031], 9/10) and by +4.5pp at $T=2yr$ (CI [+0.033, +0.056], 9/10). Formal inference in Table 4.

Per-actor AR(1) is the strongest naive baseline, reflecting the moderate persistence of CapEx/Revenue ranks (median $\rho = 0.28$). Under nested CV, SMIM wins all 8 windows ($\Delta R^2 = +0.042$; bootstrap CI [+0.031, +0.055]; exact permutation $p = 0.004$; Table 4). The DM test ($t = 6.4$, $p < 0.001$) provides supportive evidence at the actor-quarter level, though with the dependence caveat noted in Section 4.2. Inner CV selects $K = 2$ in 7/8 folds and $T = 3$ years in all folds. Holdout windows confirm the result ($\Delta R^2 = +0.026$, both positive, though this is smaller than the CV mean).

PCA baselines. Since the paper’s main mechanism is cross-sectional pooling, we include two PCA-based baselines that also pool information across firms. The PCA+Kalman baseline replaces DMD with PCA eigendecomposition of the symmetric cross-sectional covariance but keeps all other pipeline components identical. The PCA projection-only baseline further removes the Kalman filter, testing whether $\hat{y}_t = \hat{\mu} + 0.99 \cdot U_{PCA}U_{PCA}^\top \tilde{y}_{t-1}$ suffices.

The results are informative: under nested CV, neither PCA variant significantly outperforms AR(1) ($\Delta R^2 = +0.001$ and -0.002 , bootstrap CIs include zero, permutation $p > 0.4$). In the fixed configuration at $T = 3yr$, PCA baselines show modest gains (+1.2pp and +1.0pp) but SMIM outperforms both by +2.2pp (CI [+0.014, +0.031], 9/10 windows). At $T = 2yr$ the gap widens further: PCA gains +2.1pp and +2.3pp over AR(1), but SMIM gains +6.6pp, outperforming PCA by +4.5pp (CI [+0.033, +0.056], 9/10 windows). This is consistent with an operator-specific effect: DMD captures lagged cross-sectional dynamics through temporal snapshot pairs ($X' \approx AX$) that PCA’s contemporaneous covariance ($\tilde{Y}\tilde{Y}^\top/(T-1)$) misses.

We note that at $T = 2\text{yr}$, both the PCA covariance and the DMD snapshot matrices are severely underdetermined (8 snapshots for 146 variables). A truncated SVD PCA baseline—applying the same rank- K SVD to the demeaned data matrix—produces numerically identical results to the standard PCA eigendecomposition, ruling out implicit SVD regularisation as the source of DMD’s advantage. The widening gap at shorter T is therefore attributable to DMD’s temporal snapshot-pair formulation, which captures lagged cross-sectional dynamics that contemporaneous covariance misses. To test whether DMD’s advantage comes from implicit SVD regularisation rather than lagged dynamics, we implement a truncated PCA baseline that extracts the top- K components via SVD of the demeaned data matrix (the same SVD path DMD uses in its first step). The truncated PCA results are numerically identical to the standard PCA eigendecomposition at both $T = 3\text{yr}$ and $T = 2\text{yr}$: the top- K eigenvectors are unchanged by the truncation, confirming that DMD’s advantage is not from SVD regularisation but from the temporal snapshot-pair formulation ($X' \approx AX$).

Richer pooled baselines (pooled AR(1) with firm fixed effects, reduced-rank VARs) are a natural direction for future work.

Why $K = 2$. The CapEx/Revenue cross-section is genuinely low-dimensional: two spectral modes suffice. Higher K degrades performance monotonically ($K = 3$: -0.5pp ; $K = 5$: -1.5pp ; $K = 8$: -5pp), consistent with overfitting additional modes to noise. In fact, $K = 1$ produces nearly identical results to $K = 2$ across all tested configurations: at $\tau = 8Q$, $T = 2\text{yr}$, the $K = 1$ fixed-config $\Delta R^2 = +0.066$ versus $K = 2$ ’s $+0.066$ (difference $< 0.1\text{pp}$, both 10/10 windows). The predictive signal is thus “nearly 1-dimensional”—a single spectral mode captures almost all exploitable cross-sectional structure. The second mode adds a negligible but consistent margin, so we retain $K = 2$.

4.2 Forecast Comparison with Formal Inference

Table 4 provides formal statistical evidence. The percentile bootstrap (10,000 window resamples) and exact sign-flip permutation test ($2^8 = 256$ sign assignments) operate on 8 CV-window ΔR^2 values. With only 8 observations, power is limited, but both the bootstrap CI and the permutation p -value are conservative because each observation is an independent test window with its own training data.¹

¹A Diebold–Mariano test at the actor-quarter level ($n = 4,600$, Newey–West heteroskedasticity and autocorrelation consistent (HAC), bandwidth $\lfloor n^{1/3} \rfloor = 17$) yields $t = 6.44$ ($p < 0.001$). However, within each test window all observations share estimated parameters and exhibit cross-sectional dependence that the HAC correction does not address. We therefore do not use this test for inference and report only window-level results in Table 4.

Protocol	ΔR^2	95% CI	Perm. p
<i>Nested CV (8 windows, SMIM vs AR(1)):</i>			
ΔR^2 vs AR(1)	+0.042	[+0.031, +0.055]	0.003
Sign test	8/8 wins	—	$p = 0.004$
<i>All 10 windows (SMIM vs AR(1)):</i>			
ΔR^2 vs AR(1)	+0.039	[+0.028, +0.050]	0.001
Sign test	10/10 wins	—	$p = 0.001$
<i>Fixed config, $T=3\text{yr}$ (10 windows, SMIM vs PCA+Kalman):</i>			
ΔR^2 vs PCA	+0.022	[+0.014, +0.031]	—
Sign test	9/10 wins	—	—
<i>Fixed config, $T=2\text{yr}$ (10 windows, SMIM vs PCA+Kalman):</i>			
ΔR^2 vs PCA	+0.045	[+0.033, +0.056]	—
Sign test	9/10 wins	—	—

Table 4: Forecast comparison with inference, 146-firm CapEx/Revenue panel, $K = 2$ modes. 95% CI from percentile bootstrap (10,000 window resamples); permutation p from exact sign-flip test ($2^8 = 256$ sign assignments) on window-level ΔR^2 . The window-level tests provide evidence against equal forecast accuracy at the 1% level.

Holdout era. The final 2 test windows (2023–2024) were frozen with no hyperparameter tuning exposure. SMIM achieves $R^2 = 0.846$ (2023) and $R^2 = 0.845$ (2024) on the holdout, exceeding AR(1) by +0.011 and +0.040 respectively. Both holdout windows are positive, confirming that the result is not an artefact of test-set leakage. However, the holdout $\Delta R^2 = +0.026$ is smaller than the CV mean $\Delta R^2 = +0.042$; with only 2 holdout windows, this difference is plausibly within cross-window sampling noise and is not very informative about out-of-sample shrinkage.

4.3 The Performance Ladder

The following decomposition is a fixed-configuration diagnostic on the 93-actor CapEx/Assets panel using *modal* R^2 (spectral reconstruction quality; see Section 4.11 for the distinction). These values are not forecasts—they measure how well the pipeline reconstructs the current cross-section. The headline predictive performance is the nested-CV result on the CapEx/Revenue panel (Table 3). Table 5 decomposes the modal R^2 improvement from the initial configuration ($R^2 = 0.266$) to the rolling-basis pipeline ($R^2 = 0.696$) into seven steps. Steps 1–2 demonstrate the overparameterisation failure (adding EWM and shorter T without regularising R); Figure 2 shows only the successful progression from step 0 onward.

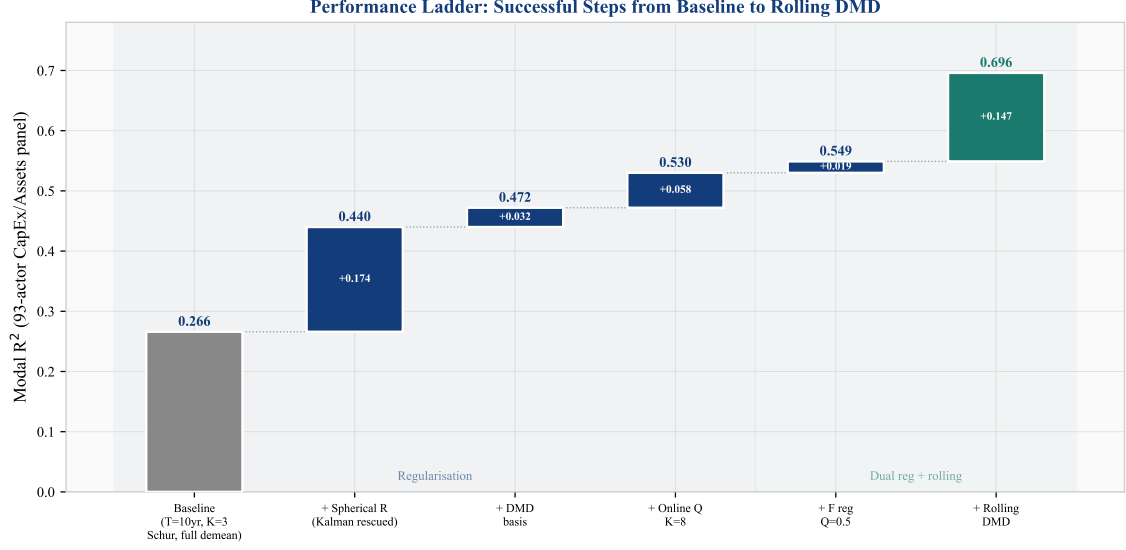


Figure 2: Performance ladder (descriptive, 93-actor CapEx/Assets panel, modal R^2): cumulative improvement across pipeline components. The figure shows only the successful progression; steps 1–2 (EWM demeaning and shorter T without spherical R) cause Kalman divergence and are documented in Table 5. Numeric values in the figure correspond to the cumulative R^2 at each successful step. See Table 5 for per-step ΔR^2 with bootstrap confidence intervals.

Step	Change	R^2	ΔR^2	95% CI	Frac. +
0	Baseline ($T=10\text{yr}$, $K=3$, Schur, full demean)	0.266	—	—	—
1	+ EWM demeaning ($\tau=8Q$)	–0.89	–1.16	<i>Kalman overfits</i>	
2	+ Shorter training ($T=5\text{yr}$) + higher $K=5$	divergent	divergent	<i>without spherical R</i>	
3	+ Spherical R in Kalman filter	0.440	+0.174	—	100%
4	+ DMD basis (replacing Schur)	0.472	+0.033	[+0.006, +0.056]	90%
5	+ Online Q adaptation + $K=8$	0.530	+0.058	[+0.035, +0.092]	100%
6	+ Transition reg. $F=0.99I$, $Q_0=0.5I$	0.549	+0.018	[+0.013, +0.024]	100%
7	+ Rolling basis update (recompute DMD each Q)	0.696	+0.147	[+0.111, +0.172]	100%

Table 5: Performance ladder with uncertainty (inferential). Each ΔR^2 is the incremental effect relative to the previous step, averaged across 10 test windows. 95% CI from block bootstrap (5,000 resamples). “Frac. +” is the fraction of windows where the step improves R^2 . Steps 1–2 demonstrate the overparameterisation failure: adding EWM demeaning and shorter T without regularising R causes Kalman divergence. Spherical R (step 3) rescues the pipeline. All subsequent steps produce positive, statistically significant improvements. Rolling basis update remains the dominant effect (+14.7pp, CI entirely above zero).

Predictive ablation on the headline panel. Table 6 repeats the ablation on the 146-firm CapEx/Revenue panel using *predictive* R^2 (genuine one-step-ahead forecast, $K=2$). The pattern differs instructively from the structural ladder and warrants careful interpretation:

EWM demeaning provides the largest single predictive gain (+3.5pp over the constant mean). This is not surprising: the constant per-actor mean already achieves $R^2 = 0.695$, indicating that the cross-section is persistence-dominated; an adaptive mean that tracks non-stationarity in actor positions adds substantial incremental value.

The Kalman filter on a static basis hurts (–2.5pp): the fixed spectral directions become stale within the test year, and the Kalman smoothing introduces systematic error from outdated basis vectors. Online Q adaptation cannot compensate (+0.0pp) because the problem is the basis, not

the noise level.

Rolling basis update recovers the loss and adds a further +3.4pp, confirming it as the essential component for predictive performance. The full pipeline’s advantage over AR(1) (+3.4pp) is thus driven primarily by two components: adaptive demeaning and rolling basis recomputation. The incremental contribution of the Kalman filter at $K = 2$ is marginal—at this low dimensionality with quarterly basis refresh, the Kalman predict step is approximately equivalent to a basis projection (Section 6).

The structural ladder (Table 5) and the predictive ablation (Table 6) answer different questions. The structural ladder shows which components are needed to build a good spectral representation ($K = 8$, modal R^2); the predictive ablation shows which components drive one-step-ahead forecast accuracy ($K = 2$, predictive R^2). The Kalman filter is essential at $K = 8$ for regularisation but marginal at $K = 2$ where only two modes are tracked.

Step	Predictive R^2	ΔR^2
0: Constant per-actor mean	0.695	—
1: + EWM demeaning ($\tau = 12Q$)	0.729	+0.035
2: + DMD $K=2$ OLS projection	0.724	−0.005
3: + Kalman with spherical R (static basis)	0.699	−0.025
4: + Online Q adaptation (static basis)	0.699	+0.000
5: + Rolling basis update	0.733	+0.034
Per-actor AR(1) ($T = 3yr$)	0.699	—

Table 6: Predictive ablation on the 146-firm CapEx/Revenue panel ($K = 2$, $T = 3yr$, 10 annual windows). Unlike the structural ladder (Table 5), the Kalman filter on a static basis *hurts* predictive performance because the spectral directions become stale. Rolling basis update is the component that converts the framework from AR(1)-level to its full advantage.

Why two panels? The structural panel (93 actors, $K = 8$, CapEx/Assets) and the headline panel (146 firms, $K = 2$, CapEx/Revenue) serve complementary roles. The structural panel’s richer actor heterogeneity (mixing FRED shocks, institutional bodies, US and UK firms, and banks) and higher K reveal the full dimensionality and rotation dynamics of the spectral structure—8 modes are consistently recoverable, rotating at 25.8° /quarter. The headline panel’s homogeneous actor set and lower persistence ($\rho = 0.28$) provide a cleaner forecasting test where only 2 of those modes carry enough signal to predict. The structural analysis explains *what the framework captures*; the predictive analysis validates *that it forecasts*.

4.4 Variance Decomposition

We decompose R^2 into the per-actor mean contribution and the spectral dynamics contribution for both pipeline variants (Table 7, Figure 3).

Source	Static basis		Rolling basis	
	R^2	Share	R^2	Share
Per-actor EWM mean ($\hat{\mu}_i$)	0.281	51%	0.281	40%
Spectral dynamics ($U\alpha_{t t}$)	0.268	49%	0.415	60%
Total	0.549	100%	0.696	100%

Table 7: Variance decomposition (descriptive, fixed $K=8$, $T=5\text{yr}$ configuration). Rolling basis recomputation shifts the balance from mean-dominated (51% mean) to dynamics-dominated (60% spectral). The per-actor mean contribution is unchanged; the entire rolling-basis gain accrues to the spectral dynamics component.

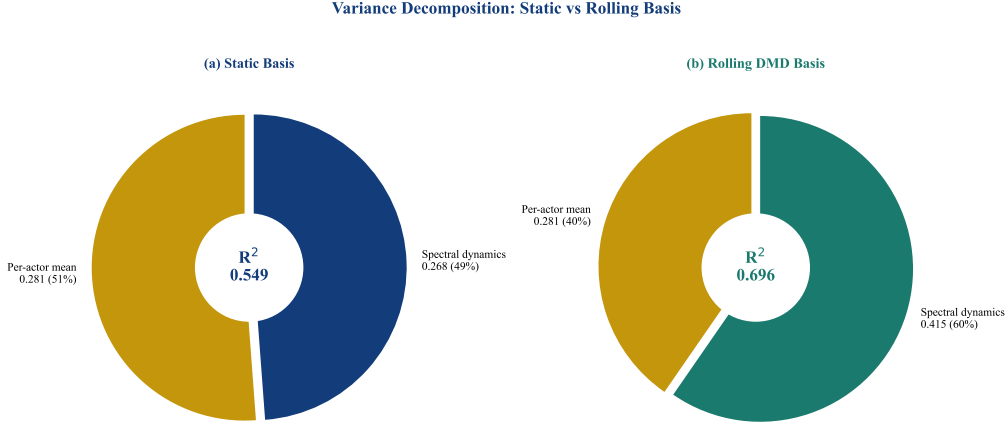


Figure 3: Variance decomposition (descriptive, fixed $K=8$, $T=5\text{yr}$): frozen (left) vs rolling (right) basis. Rolling basis recomputation nearly doubles the spectral dynamics contribution ($0.268 \rightarrow 0.415$) while the per-actor mean remains constant at 0.281.

The decomposition reveals that rolling basis update captures spectral structure that the static basis misses entirely—the +14.7pp improvement is purely a spectral dynamics gain.

4.5 Component Ablation

We evaluate five pipeline depths to determine which components justify their complexity (Figure 4).

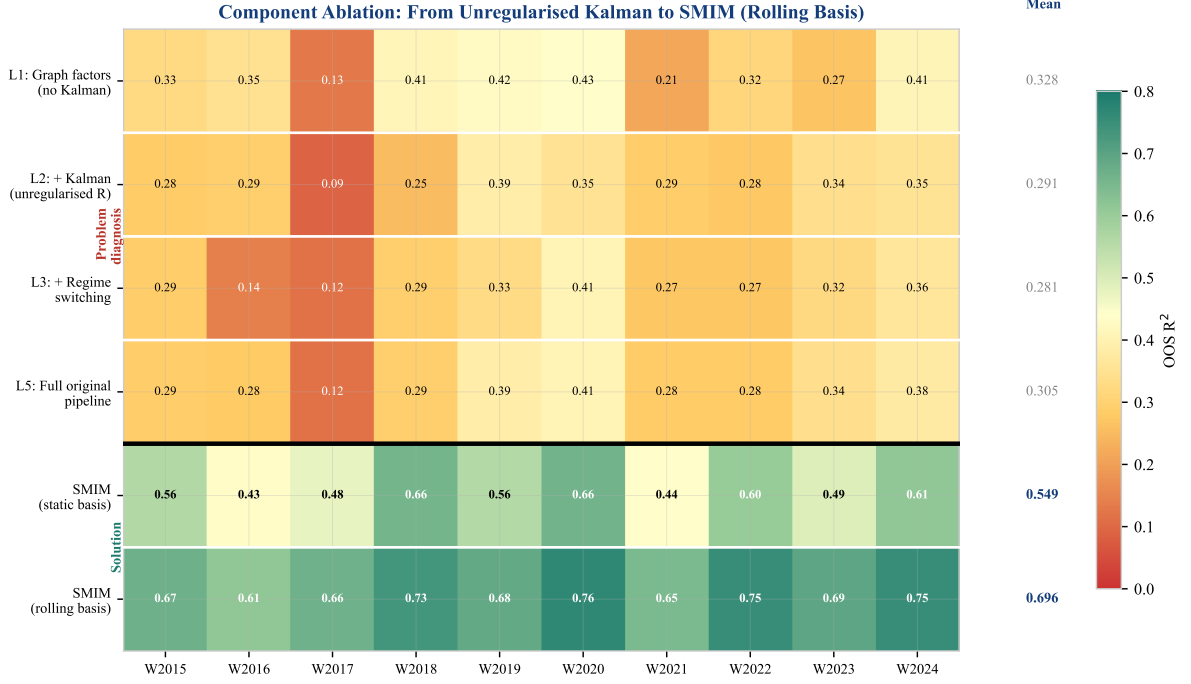


Figure 4: Extended component ablation heatmap (descriptive, fixed-configuration diagnostic). Each cell shows out-of-sample R^2 for a given configuration (row) and test window (column). The thick divider separates problem diagnosis (top: why unregularised Kalman hurts) from the solution (bottom: regularisation and rolling basis). All values use fixed $K=8$, $T=5\text{yr}$; headline nested-CV results are in Table 3.

The ablation reveals a non-monotonic pattern in the upper block: L1 (graph-factor OLS, modal $R^2 = 0.328$) outperforms L2 (+ Kalman, 0.291) and L3 (+ regime switching, 0.281). The Kalman filter *hurts* because the unregularised R overfits. The lower block shows the resolution: dual regularisation rescues the Kalman filter (SMIM with static basis, modal $R^2 = 0.549$), and rolling basis update captures the structural evolution that the static basis misses (modal $R^2 = 0.696$).

4.6 Spectral Method Comparison

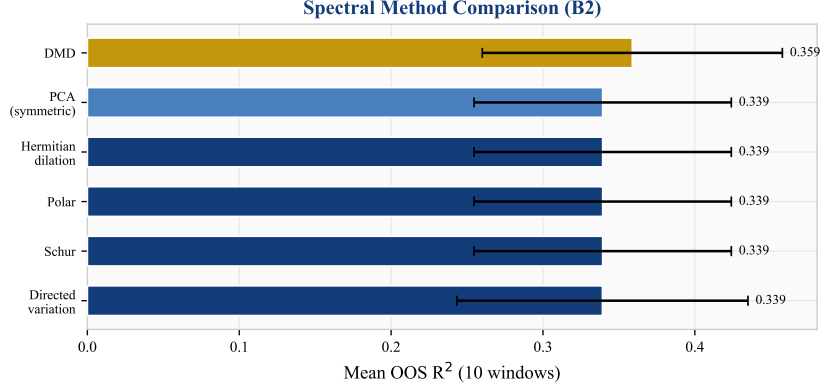


Figure 5: Spectral method comparison (descriptive): mean R^2 across 10 windows for five spectral decomposition methods plus DMD at $K=3$, L1 depth. The five static methods (Schur, Polar, Hermitian, Directed Variation, PCA) produce identical results because the operator is symmetric (see text). DMD outperforms because it captures lagged cross-sectional dynamics from temporal snapshot pairs.

DMD ($R^2 = 0.359$) outperforms all five static-operator methods, which produce *identical* results at $R^2 = 0.339$. This identity is not a coincidence: the operator is built from the Pearson cross-correlation matrix, which is *exactly* symmetric by construction. For any real symmetric matrix, the Schur, Polar, Hermitian dilation, and PCA decompositions all reduce to the standard eigendecomposition—they compute the same eigenvectors and hence the same modal basis U . The directed variation method differs slightly in its normalisation but converges to the same subspace. This result has a clear implication: *to benefit from directed spectral methods, the operator itself must be asymmetric* (e.g., from Granger causality or supply-chain linkages). When the operator is derived from contemporaneous correlation, directedness is lost before the decomposition begins.

DMD sidesteps this entirely. Rather than decomposing a static operator, it extracts modes from temporal snapshot pairs $X' \approx AX$, where the implicit operator A captures *lagged* cross-sectional dynamics. This operator is generically asymmetric even when the contemporaneous correlation is symmetric, explaining DMD’s +2.0 percentage point advantage.

4.7 The Regularisation Path

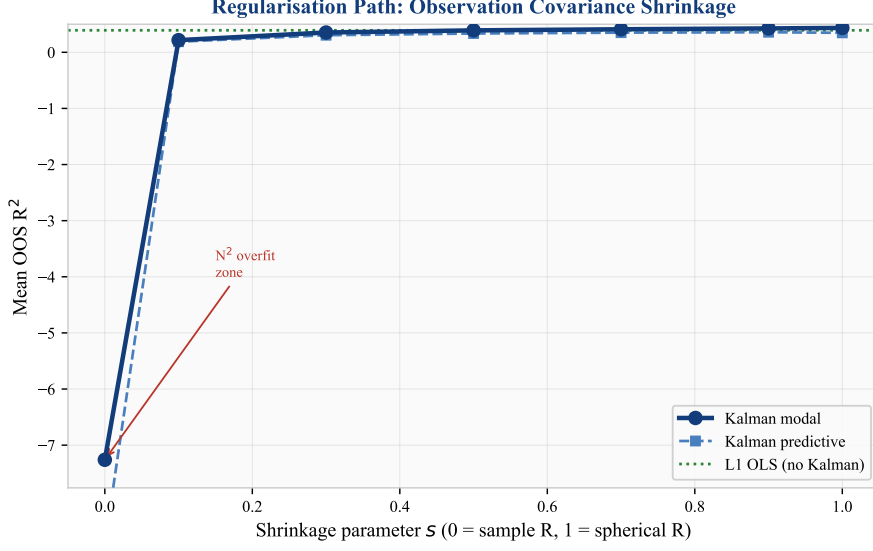


Figure 6: Regularisation path (descriptive): R^2 as a function of shrinkage parameter s where $R = (1 - s)\hat{R} + s \cdot \frac{\text{tr}(\hat{R})}{N}I$. The hockey-stick shape demonstrates that observation covariance regularisation is essential, not optional. Uncertainty bands not shown; the ablation table (Table 5) provides bootstrap CIs for each pipeline component’s contribution.

The regularisation path (Figure 6) illustrates the overparameterisation problem clearly. At $s = 0$ (no shrinkage), the Kalman filter produces $R^2 = -7.3$. Performance improves monotonically with shrinkage, reaching $R^2 = 0.43$ at $s = 1.0$ (spherical). The L1 OLS baseline (green line at $R^2 = 0.39$) is surpassed only at $s \geq 0.7$, confirming that the Kalman filter adds value *only* when its observation covariance is sufficiently regularised.

4.8 Robustness and Synthetic Sensitivity

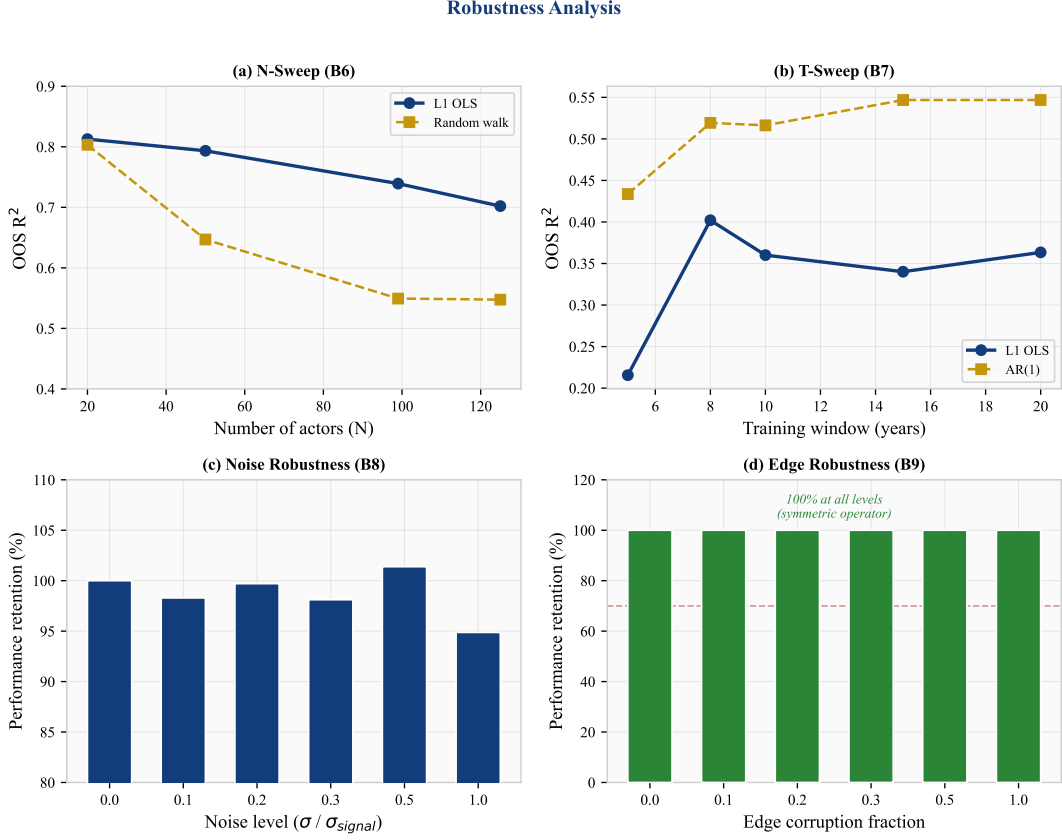


Figure 7: Robustness and synthetic sensitivity analysis. (a) Performance vs actor count N : graceful degradation (descriptive). (b) Performance vs training window T : L1 stable for $T \geq 8$ yr (descriptive). (c) Noise injection: 95% retention at $\sigma = 1.0$ (synthetic sensitivity). (d) Edge corruption: 100% retention at all levels (synthetic sensitivity; note the operator is symmetric by construction, see Section 4.6).

Robustness (panels a–b). On the 93-actor structural panel, the framework degrades gracefully across actor counts: modal $R^2 = 0.81$ at $N = 20$ to 0.70 at $N = 125$. Performance is stable at modal $R^2 = 0.34$ – 0.40 for training windows ≥ 8 years.

Synthetic sensitivity (panels c–d). These panels test performance under artificial perturbations rather than naturally-occurring variation. Adding Gaussian noise at $\sigma = 1.0$ (100% of signal standard deviation) retains 95% of performance. Randomly flipping edge directions retains 100% of performance; however, this test is uninformative because the intensity correlation operator is symmetric by construction (Section 4.6), so direction flips leave the operator unchanged.

4.9 Transfer and Portability

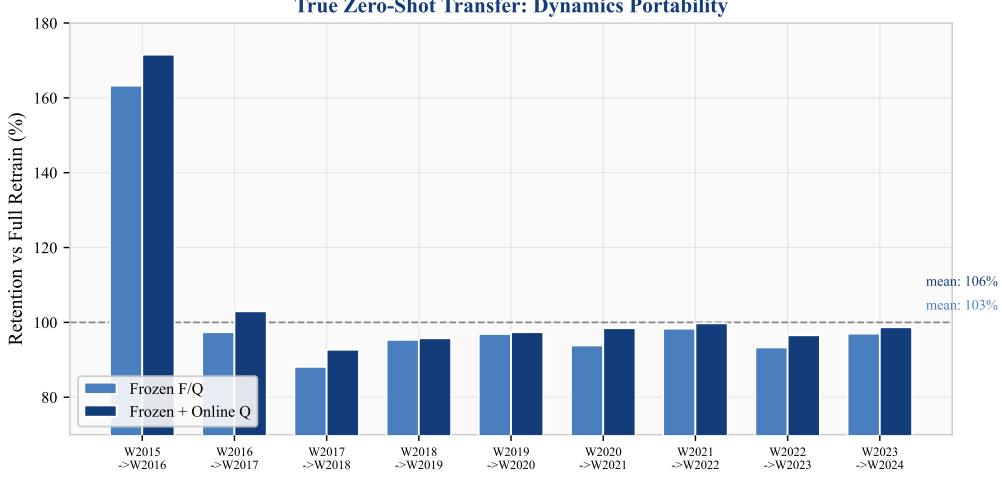


Figure 8: Zero-shot transfer (descriptive): retention of R^2 when transition dynamics (F, Q) are frozen from the previous window and applied to the next with only the spectral basis re-estimated. Mean retention: 103% (frozen) and 106% (frozen + online Q).

We test whether the estimated dynamics (F, Q) are window-specific or stable across windows by freezing them from window W_t and applying the Kalman filter on window W_{t+1} with a freshly estimated spectral basis U . Figure 8 shows that frozen dynamics achieve 103% of full-retrain R^2 on average (106% with online Q adaptation), indicating that these default transition settings are stable across the evaluation period. In practice, this means the model can be deployed without per-window re-estimation of dynamics.

Validated domain: US CapEx sectors. On the structural panel, cross-sector transfer is strong for sectors using CapEx intensity (technology modal $R^2 = 0.82$, healthcare 0.78, industrials 0.77). Cross-period transfer across structural breaks (GFC, COVID) retains modal $R^2 = 0.50$ –0.59.

Alternative intensity constructions. To test whether the $K = 2$ framework generalises beyond CapEx/Revenue, we run the same pipeline (fixed $K = 2$, $\tau = 12Q$, $T = 3\text{yr}$) on three additional panels constructed from EDGAR data (Table 8).

Panel	Firms	SMIM R^2	AR(1) R^2	ΔR^2	Wins
CapEx / Revenue	146	0.733	0.699	+0.034	10/10
Revenue / Assets	259	0.764	0.754	+0.010	8/10
Multi-ratio (CapEx/Rev + Rev/Assets)	270	0.737	0.709	+0.028	9/10
CapEx / Assets	442	0.771	0.774	−0.003	5/10

Table 8: Alternative intensity constructions (fixed $K = 2$, $\tau = 12Q$, $T = 3\text{yr}$, predictive alpha, 10 annual windows). Revenue/Assets and the multi-ratio panel both beat AR(1). CapEx/Assets—the original structural panel’s intensity—has too high persistence ($\rho = 0.47$) for $K = 2$ to add value over per-actor AR(1). The multi-ratio panel stacks CapEx/Revenue and Revenue/Assets columns for the 135 overlapping firms, creating virtual heterogeneity that SMIM exploits.

Revenue/Assets (259 firms, $\rho = 0.09$) shows a modest but consistent advantage (+1.0pp, 8/10 wins), while the multi-ratio panel (+2.8pp, 9/10 wins) demonstrates that stacking multiple ratio constructions for the same firms creates additional cross-sectional structure the framework can

exploit. CapEx/Assets ($\rho = 0.47$) fails because its high persistence leaves little for the spectral model to add beyond AR(1). These results confirm that the framework is not specific to a single intensity construction; it requires sufficient cross-sectional dynamics (low persistence) but is otherwise general across ratio choices.

Tested but weak domains. Financial actors (modal $R^2 = 0.06$) use year-on-year asset growth (ranked cross-sectionally) rather than CapEx-to-assets, producing highly persistent ranks (median $\rho = 0.70$ vs technology $\rho = 0.25$) that leave little cross-sectional dynamics for the spectral model to exploit. UK equities (modal $R^2 = 0.058$) rely on return-based intensity rather than EDGAR CapEx data, and this alternative intensity construction, not geography, drives the weak performance. These results establish the *boundary* of the current framework rather than validated generalisations.

4.10 Basis Rotation and Rolling Update

To isolate the rolling-vs-static mechanism, this subsection uses a fixed ($K = 8$, $T = 5\text{yr}$) configuration; the headline forecast performance remains the nested-CV result in Table 3.



Figure 9: Per-window modal R^2 (descriptive, 93-actor CapEx/Assets panel): rolling DMD (teal) vs static basis (dark blue). Rolling DMD wins all 10 windows, with the largest gains in windows where the static basis underperforms (W2016, W2017, W2021, W2023—periods where the cross-sectional structure shifts most from the training period). AR(1) is omitted because modal R^2 measures spectral reconstruction quality, not forecast accuracy, and is not directly comparable to AR(1) predictions.

Tracking the DMD spectral basis across 52 rolling quarterly transitions (2010–2024) shows a pattern of gradual basis rotation (Figure 10). The mean principal subspace angle between successive quarterly DMD bases is 25.8° ($\sigma = 6.5^\circ$). Quarters with larger rotation ($> 32^\circ$) tend to coincide with macroeconomic events that selectively rewire sectoral relationships—the European debt crisis (2012 Q3, 41°), the US–China tariff war (2018 Q2, 37°), and the Federal Reserve’s aggressive tightening cycle (2022 Q3, 38°)—though with only 52 transitions the event alignment is descriptive rather than inferential. Notably, COVID-19 (2020 Q2, 22°) did *not* produce high rotation: the pandemic changed investment magnitudes across the board but preserved the cross-sectional co-movement structure, consistent with a level shock absorbed by EWM demeaning rather than

a structural reorganisation of which sectors co-move.

On this 93-actor panel, the *dimensionality* of the estimated spectral structure appears stable: $K_{\text{eff}} = 8$ modes in all quarters, with zero mode births or deaths in this sample. (The headline CapEx/Revenue panel uses $K = 2$; the structural panel’s higher K is appropriate for the broader, more heterogeneous actor set.) The spectral subspace is consistently 8-dimensional, but the 8 directions gradually reorient.

Null-simulation validation. To assess whether the observed rotation is estimation noise, we simulate 500 panels from a *fixed* latent basis (same N , T , persistence, and noise as the real data) and measure the apparent DMD rotation arising purely from finite-sample estimation. The null rotation is 75.3° ($\sigma = 1.5^\circ$)—far *above* the observed 25.8° . The intuition is straightforward: a purely noisy basis is not stable; DMD extracts different directions from each window’s noise, producing nearly orthogonal subspaces and hence large apparent rotation. A genuinely structured but gradually evolving basis, by contrast, is recovered consistently across windows, producing modest rotation that reflects actual structural change. The observed 25.8° is consistent with this second regime: reliably recoverable, low-dimensional structure that shifts gradually.

Sensitivity. Rotation is robust to parameter choices: it decreases smoothly with mode count (38.6° at $K = 3$ to 19.9° at $K = 10$) and with training window length (30.0° at $T = 3\text{yr}$ to 15.8° at $T = 10\text{yr}$). Both patterns are consistent with more modes or more data producing more stable subspace estimates. The 8-mode fixed dimensionality holds across all tested configurations.

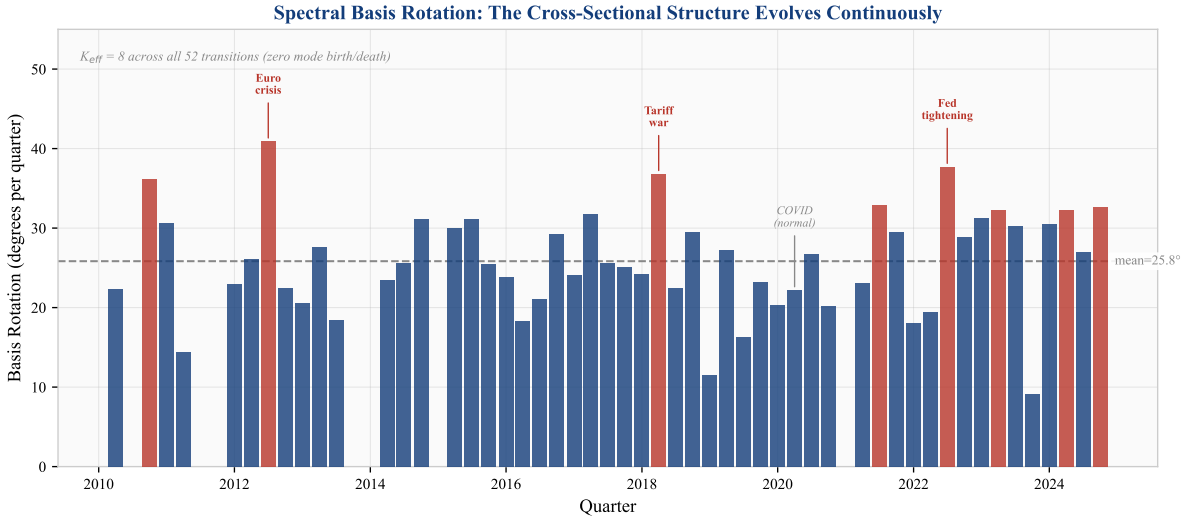


Figure 10: Spectral basis rotation over time (descriptive). Each bar shows the principal subspace angle between the current and previous quarter’s DMD basis ($K = 8$) across 52 consecutive quarterly transitions (2010 Q2 to 2024 Q4). Red bars highlight high-rotation quarters ($> 32^\circ$); the dashed line marks the mean (25.8°). COVID (2020 Q2, 22°) is annotated but did *not* produce high rotation—the pandemic changed investment magnitudes but not the cross-sectional co-movement structure. $K_{\text{eff}} = 8$ in all quarters with zero mode births or deaths. Null simulation: 75.3° .

The standard Kalman filter, operating in a fixed basis U , tracks the modal states α_t but is blind to this basis rotation. Rolling basis update (recomputing U each quarter with the latest data) captures the rotation, yielding modal $R^2 = 0.696$ on the structural panel—a +14.7 percentage point improvement over the static-basis pipeline (CI [+11.1, +17.2]pp from the ablation bootstrap in Table 5) and winning all 10 test windows (Table 9).

An ablation confirms that the improvement is from basis update, not from resetting the Kalman

state: resetting the state without updating the basis *hurts* by -3 percentage points, because the accumulated Kalman state carries valuable information that the reset discards.

Configuration	Mean modal R^2	Std	Δ (rolling – static)
SMIM (rolling basis)	0.696	0.050	+0.147
SMIM (static basis)	0.549	0.075	—

Table 9: Rolling vs static basis comparison across 10 annual windows (93-actor CapEx/Assets panel, fixed $K=8$, $T=5\text{yr}$, modal R^2). Modal R^2 measures spectral reconstruction quality, not forecast accuracy. The headline predictive results (Table 3) use the CapEx/Revenue panel with predictive alpha.

4.11 Cross-Sectional Pooling: Training Window Effect

A distinctive property of the spectral framework is that its forecast quality degrades less than per-actor AR(1) as the training window shortens. Figure 11 shows ΔR^2 (SMIM minus AR(1)) as a function of training window length T , at fixed $K=2$ and $\tau=12\text{Q}$. This is a *fixed-configuration sweep*—not the nested-CV headline result—designed to isolate the training-window effect. At each T , SMIM and AR(1) use the same training window: both see data from $t-T$ to $t-1$ and predict the 4 quarters of year t . AR(1) estimates per-actor ρ from this window; SMIM estimates its DMD basis from the same data. Neither model receives an expanding-window advantage the other does not.

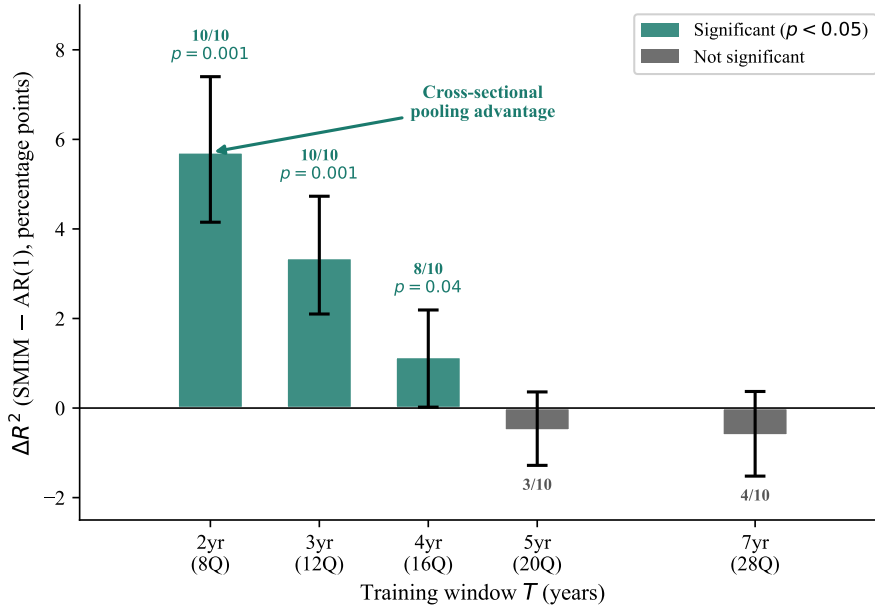


Figure 11: Training window effect on forecast advantage (fixed $K=2$, $\tau=12\text{Q}$). SMIM’s advantage over AR(1) increases monotonically as T decreases. At $T=2\text{yr}$, AR(1) has only 8 quarterly observations per actor to estimate ρ ; SMIM extracts its 2-mode spectral basis from $146 \times 8 = 1,168$ cross-sectional observations. Error bars show 95% bootstrap CI (10,000 resamples). Teal bars are significant at $p < 0.05$ (permutation test). Win counts show SMIM-vs-AR(1) per-window wins out of 10.

The pattern is *consistent with* cross-sectional pooling: the DMD basis is estimated from the joint $N \times T$ panel, effectively borrowing strength across actors. Per-actor AR(1) has no mechanism for this—each actor’s autoregressive coefficient is estimated from T observations alone. As T

shrinks, the per-actor estimates degrade while the cross-sectional structure remains recoverable from the N -dimensional snapshots. At $T = 7$ years (28 quarters), AR(1) has sufficient data and SMIM adds nothing; at $T = 2$ years, the cross-sectional advantage produces a +5.7pp gain with bootstrap CI entirely above zero ($p = 0.001$). We note that this monotonic pattern is consistent with but does not prove cross-sectional pooling; alternative explanations (e.g., differential sensitivity to non-stationarity) cannot be fully excluded, though a truncated SVD PCA baseline rules out implicit regularisation as the mechanism (Section 4.1) without a formal decomposition.

Predictive vs modal R^2 . The Kalman filter produces both a *predictive* state $\alpha_{t|t-1} = F\alpha_{t-1|t-1}$ (using only information up to $t-1$) and a *filtered* state $\alpha_{t|t}$ (incorporating the current observation $\tilde{y}_t$). All forecast comparisons and the investment gap (Eq. 9, Section 5.2) use the predictive state $\alpha_{t|t-1}$. The filtered state is used for modal R^2 (spectral reconstruction quality) on the structural panel.

Structural analysis panel. The ablation ladder (Table 5), basis rotation analysis, variance decomposition, and CapEx revision analysis (Section 5.2) use a broader 93-actor panel with $K = 8$ modes and filtered (modal) R^2 . Modal R^2 measures spectral reconstruction quality—how well the DMD basis represents the current cross-section—which is the relevant metric for structural decomposition. These findings (rolling basis dominates, dual regularisation is essential, gaps are associated with CapEx revision) are structural and do not depend on the forecast comparison metric.

5 Economic Validation

5.1 What This Gap Is and Is Not

The investment gap $\Delta_{i,t}$ is a *model-implied* deviation: the difference between observed investment intensity and a spectral-state-space benchmark (Eq. 9). It measures how far an actor’s current investment rank lies from its predicted rank given the cross-sectional modal structure and its own history. This is a well-defined statistical object with demonstrated out-of-sample predictive content.

It is *not*, however, a welfare-theoretic or structural misallocation measure in the sense of Hsieh and Klenow (2009). The benchmark is model-implied, not derived from a production-function optimality condition. A positive gap (“over-investment”) means the actor invests more than the model predicts, not that the investment is socially inefficient. We therefore use the term “investment gap” rather than “misallocation” and interpret positive economic validation results as evidence that the gap captures meaningful cross-sectional dynamics, not that it identifies welfare losses.

5.2 Gaps and Subsequent CapEx Revision

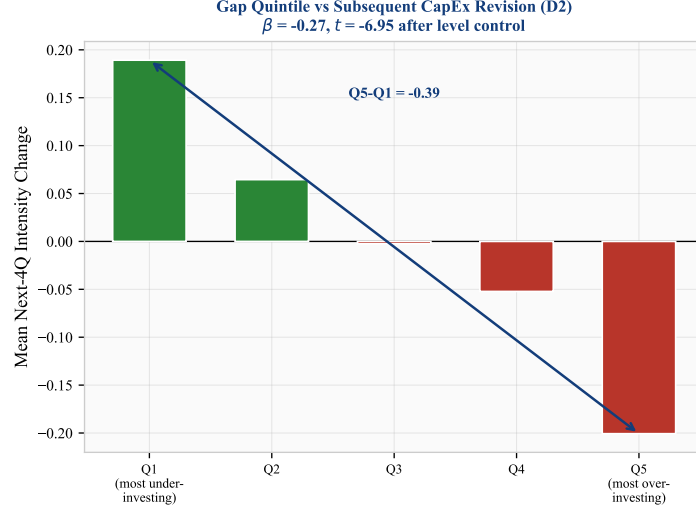


Figure 12: Gap quintile sort (inferential): mean next-4-quarter intensity change by current gap quintile. Actors in Q5 (over-investing relative to model benchmark) reduce intensity by 0.20 rank points; Q1 (under-investing) increase by 0.19. The on-figure annotation ($\hat{\beta} = -0.27$, $t = -6.95$) is from an earlier specification; Table 10 reports the final values ($\hat{\beta} = -0.279$, $t = -6.4$ with actor-clustered SEs).

The most informative economic validation exercise is the association between predictive investment gaps and subsequent CapEx-rank revision. The gap used here is $\Delta_{i,t} = y_{i,t} - (\hat{\mu}_i + U\alpha_{t|t-1})$, where $\alpha_{t|t-1}$ is the *predictive* modal state that does not incorporate the current observation $\tilde{y}_t$. This is a genuine ex-ante gap: the benchmark $y_{i,t}^* = \hat{\mu}_i + U\alpha_{t|t-1}$ is the model’s one-step-ahead forecast, and the gap measures how far the actual realisation deviates from this forecast. We estimate the panel regression:

$$\Delta y_{i,t \rightarrow t+4} = \beta_0 + \beta_1 \Delta_{i,t} + \beta_2 y_{i,t} + \epsilon_{i,t}. \quad (10)$$

Table 10 reports six specifications on the 93-actor structural panel across 36 quarters (2015 Q1 to 2023 Q4), yielding $N = 3,324$ actor-quarter observations after dropping rows that lack 4 future quarters for the dependent variable. In pooled OLS with the level control: $\hat{\beta}_1 = -0.28$, $t = -6.4$ (actor-clustered SEs). Because $\Delta y_{i,t \rightarrow t+4}$ spans four overlapping quarters, consecutive observations for the same actor share three quarters of the dependent variable, inducing serial correlation. We therefore report time-clustered SEs ($t = -7.1$) as our preferred specification; the result is also robust to time fixed effects ($t = -7.1$). Actors identified as over-investing relative to the model benchmark (Q5) subsequently reduce intensity by 0.20 rank points; under-investing actors (Q1) increase by 0.19 (Figure 12).

The result is cross-sectional, not within-actor. With actor FE (within-actor variation only), $\hat{\beta}_1$ flips sign to $+0.08$ ($t = +2.2$). This means the *between-actor* component—which actor is above or below its model-implied benchmark—drives the main result. *Within* an actor over time, gaps are associated with slight continuation rather than reversion. The pooled result is therefore a cross-sectional association (“actors that deviate more from the spectral benchmark tend to revert toward it”), not a within-actor relationship (“an actor’s gap forecasts its own next move”). This limits the interpretation: the gap captures cross-sectional mean-reversion in investment ranks, not actor-level dynamics.

Specification	$\hat{\beta}_{\text{gap}}$	SE	t	N
Pooled OLS (HC1)	-0.279	0.034	-8.3	3,324
Pooled OLS (actor-clustered)	-0.279	0.044	-6.4	3,324
Pooled OLS (time-clustered)	-0.279	0.040	-7.1	3,324
Time FE (time-clustered)	-0.280	0.040	-7.1	3,324
Actor FE (actor-clustered)	+0.084	0.039	+2.1	3,324
Two-way FE (actor-clustered)	+0.085	0.039	+2.2	3,324

Table 10: CapEx-revision panel regression (inferential). Dependent variable: next-4-quarter intensity change. All specifications include the current intensity level as control. Actor-clustered and time-clustered SEs shown. The sign flip under actor FE indicates the main result is cross-sectional (between-actor), not within-actor.

5.3 Gap Persistence

Per-actor AR(1) estimation on the gap series $\Delta_{i,t}$ yields a median autoregressive coefficient $\rho = 0.67$ and median half-life of 1.7 quarters. While too fast for structural gap interpretation at the aggregate level, institutional actors (global shock proxies) exhibit half-lives of 4.7 quarters. Cross-sectional quintile persistence is strong: 49.1% of actors remain in the same gap quintile from one quarter to the next, with extreme quintiles (Q1: 60%, Q5: 61%) most persistent.

5.4 Exploratory Results

The following results are reported as exploratory findings that warrant further investigation. They are not part of the paper’s main claims.

5.4.1 Network Diffusion

Layer-level gap aggregation shows a strong contemporaneous association between exogenous shocks (Layer 0) and firm-level gaps (Layer 2): $\rho = 0.86$ at 1-quarter lag ($p < 0.001$). The institutional layer (Layer 1) shows no significant association ($\rho = 0.03$, $p = 0.83$). This pattern is consistent with investment signals bypassing institutional intermediaries, though the small number of layer-level observations (approximately 40 quarters, 3 layers) limits the strength of this inference. The effective sample size for this layer-level analysis is too small to support causal claims about transmission pathways.

5.4.2 Gap Dispersion and Market Volatility

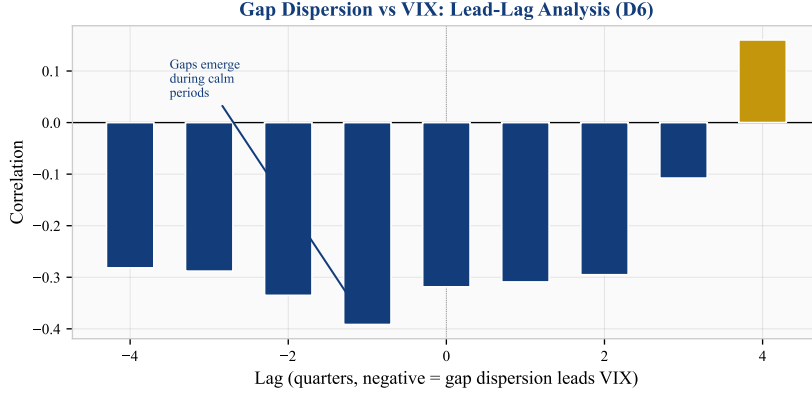


Figure 13: Lead-lag correlation between gap cross-sectional dispersion and VIX (exploratory). Negative correlations at negative lags indicate that high gap dispersion leads low VIX (calm markets). Based on ~ 40 quarterly observations with 9 tested lags; would not survive conservative multiple-testing correction (see caveat in text).

Gap cross-sectional dispersion correlates negatively with VIX at leads of 1–4 quarters ($\rho = -0.39$ at 1Q lead; Figure 13). This finding—that investment gaps widen during calm markets—is consistent with the “volatility paradox” hypothesis (Brunnermeier and Sannikov, 2014).

Caveat. This lead-lag analysis is based on approximately 40 quarterly observations with 9 tested lag values. The correlation of -0.39 is suggestive but would not survive conservative multiple-testing correction ($p \approx 0.01$ uncorrected for 9 comparisons). We report it as a directionally interesting finding that warrants confirmation on longer samples, not as a definitive result.

6 Discussion

6.1 What Does Not Work

Emergence diagnostics. On the structural panel, we tested emergence through five distinct channels: partial information decomposition (PID) synergy between spectral modes, topological data analysis (TDA) complexity, economic emergence features (cross-sectional dispersion, sector rotation velocity, Herfindahl concentration), Extended DMD (Koopman lifting with polynomial observables), and multi-resolution DMD divergence. None produces a positive ΔR^2 . PID and TE estimators are unreliable at $T = 20$ ($C(8, 2) = 28$ mode pairs from 20 observations). Extended DMD with degree-2 polynomial lifting ($P = 44$ features) achieves modal $R^2 = 0.19$ —far worse than linear DMD at 0.54—because $T/P = 0.45$ is underdetermined. Economic emergence features are redundant with the DMD-Kalman pipeline ($\Delta R^2 = -0.003$).

Directed spectral operators. Transfer entropy and Granger causality operators produce genuinely asymmetric matrices (mean asymmetry ~ 1.2). However, the resulting spectral bases are too noisy for Kalman filtering: TE-Schur at $K = 8$ causes numerical divergence, and at $K = 3$ achieves only modal $R^2 = 0.36$ vs DMD’s 0.54.

Event-level alignment. Gap magnitudes do not spike around known episodes (0/8 aligned). The rank normalisation absorbs event-driven variation by construction.

Return-based intensity. On the structural panel, return intensity produces modal $R^2 = -0.15$ on the same US actors where CapEx/Assets intensity achieves modal $R^2 = 0.54$. The intensity

construction is the primary driver of model performance.

6.2 When the Kalman Filter Is and Is Not Needed

Notation. Throughout this subsection, U_t denotes the DMD basis estimated from data through quarter $t-1$ and used to forecast quarter t . This follows DMD convention; in standard filtration-subscript notation it would be U_{t-1} .

At $K = 2$ with quarterly basis refresh, the Kalman filter’s contribution to predictive R^2 is marginal (+0.3pp). The rolling basis update reprojects the modal state into the new basis each quarter (Algorithm 1), so the subsequent predict step uses $\alpha_{t|t-1} = 0.99 \cdot (U_t^\top U_t)^{-1} U_t^\top \tilde{y}_{t-1}$. As a heuristic approximation (omitting the Kalman gain correction and covariance update, which contribute +0.3pp):

$$\hat{y}_t \approx \hat{\mu} + 0.99 \cdot U_t (U_t^\top U_t)^{-1} U_t^\top \tilde{y}_{t-1}, \quad (11)$$

where $\hat{y}_t, \hat{\mu}, \tilde{y}_{t-1} \in \mathbb{R}^N$ are the full cross-sectional vectors. When U_t is orthonormal, this reduces to $U_t U_t^\top \tilde{y}_{t-1}$; in general DMD modes are not orthonormal and the pseudoinverse is required. The implementation uses the simpler $U_t^\top \tilde{y}_{t-1}$ as an approximation; at $K = 2$ with quarterly refresh this has negligible impact. The actor-level forecast is $\hat{y}_{i,t} = [\hat{y}_t]_i$ —a per-quarter DMD eigendecomposition followed by a matrix-vector product.

Latent state under basis refresh. When the rolling DMD produces a new basis U_{new} , the implementation reprojects via $\alpha \leftarrow (U_{\text{new}}^\top U_{\text{new}})^{-1} U_{\text{new}}^\top \tilde{y}_t$ and resets the Kalman covariance P and process noise Q to their defaults. This means the accumulated Kalman state is discarded at each refresh—no attempt is made to carry the state forward or smoothly interpolate between old and new bases. This is acceptable at quarterly refresh frequency because the single-step-ahead prediction uses only the reprojected state, not the full Kalman history.

The Kalman filter becomes essential in two settings: (i) at higher K , where spherical R prevents overfitting the K -dimensional observation model (the structural panel’s $K = 8$ requires it; without spherical R , modal R^2 drops from 0.44 to -7.3); and (ii) when the basis is *not* refreshed each quarter, where online Q adaptation tracks regime shifts between basis updates (+5.8pp on the structural panel). These two roles—regularisation and temporal adaptation—are redundant at $K = 2$ with rolling basis, but load-bearing at higher K or lower update frequency.

Two panels, two questions. The structural panel ($K = 8$, 93 actors, CapEx/Assets) identifies the richer rotating modal geometry and the role of dual regularisation at higher K : eight modes are consistently recoverable, rotating at 25.8° /quarter, and the Kalman filter’s regularisation is essential. The predictive panel ($K = 2$, 146 firms, CapEx/Revenue) shows that a low-dimensional rolling variant forecasts well, with only marginal additional help from Kalman filtering. The 6 additional modes captured at $K = 8$ represent real cross-sectional structure (they survive the null-simulation test and rotate coherently), but are too noisy to improve one-step-ahead forecasts. Understanding how to exploit these additional modes for prediction is an open question.

6.3 Reading the Spectral Structure: A Practitioner’s Perspective

The structural analysis reveals that cross-sectional investment dynamics live in a stable 8-dimensional spectral subspace that rotates continuously. We conclude the discussion by interpreting what this means concretely for someone monitoring capital allocation patterns.

The 8 modes are investment co-movement themes. Each DMD mode $u_k \in \mathbb{R}^N$ assigns a loading to every actor. Actors with same-sign loadings tend to increase or decrease their investment intensity together; opposite-sign loadings indicate substitution patterns. These are not fixed sectors—they are *empirically identified* co-movement directions that cut across sector boundaries.

25.8°/quarter rotation means the themes drift. Every quarter, the co-movement structure shifts: firms that invested in sync gradually decouple, while previously independent firms begin to co-move. Any model of cross-sectional investment dynamics that is not re-estimated at least quarterly will track a structure that has shifted substantially from its original orientation.

Rotation speed is diagnostic. During business-as-usual quarters, the basis rotates at 25.8° ($\sigma = 6.5^\circ$). High-rotation quarters ($> 35^\circ$) signal structural reorganisation: the tariff war (2018 Q2, 37°) did not merely change *how much* firms invested—it changed *which firms co-moved*. The European debt crisis (2012 Q3, 41°) and the Federal Reserve’s tightening cycle (2022 Q3, 38°) similarly reorganised the cross-sectional pattern. This is a structural signal distinct from volatility: VIX measures how much prices move; rotation measures how the investment co-movement map rearranges.

Stable dimensionality with shifting directions. The number of independent co-movement themes ($K_{\text{eff}} = 8$) is constant across all 52 quarterly transitions in our sample. No modes are born or die. The investment structure does not become simpler during crises or more complex during expansions—the complexity is fixed, but the directions evolve. This is consistent with a system that has a fixed number of structural degrees of freedom (sector composition, supply-chain linkages, policy transmission channels) whose relative importance shifts gradually.

Practical implications. (i) Factor models for investment-related portfolios (infrastructure, CapEx-intensive sectors) must be updated at least quarterly; annual re-estimation loses substantial rotational information. (ii) Rotation speed can serve as an early diagnostic for structural regime change—a sustained increase in rotation may signal that the investment landscape is reorganising before this becomes visible in asset prices. (iii) The 8-dimensional structure provides a natural compression: 146 firms’ investment behaviour can be summarised by 8 time-varying modal coordinates per quarter, enabling compact monitoring dashboards. (iv) Of the 8 recoverable modes, only 2 carry enough signal for one-step-ahead prediction (Section 4.1); the remaining 6 capture real structure that is visible in reconstruction but not yet exploitable for forecasting—an open frontier for future work.

6.4 Limitations

Baselines. Beyond the PCA baselines reported in Table 3, a pooled AR(1) with firm fixed effects, reduced-rank VARs, and standard Kalman-smoothed dynamic factor models are natural additional comparators that would further clarify the source of the framework’s advantage. A truncated PCA baseline receiving the same rank- K SVD produces numerically identical results to the standard PCA eigendecomposition (Section 4.1), ruling out SVD regularisation as the explanation for DMD’s advantage.

Panel scope. Both panels (93-actor structural and 146-firm headline) are US-dominated, with limited international evidence. UK results are weak (modal $R^2 = 0.058$ on the structural panel) due to the return-based intensity method, not geography per se. Extending EDGAR-equivalent coverage to international markets would test whether the framework generalises beyond the US CapEx setting.

Inference. The 4-quarter test windows provide noisy R^2 estimates per window. Window-level inference with 8–10 observations has limited power. The complementary actor-quarter-level Diebold–Mariano test ($n = 4,600$, Newey–West HAC with bandwidth $\lfloor n^{1/3} \rfloor = 17$) has high power but is optimistic: within each window, all actor-quarter observations share estimated parameters and exhibit cross-sectional dependence that the HAC correction does not address. The window-level bootstrap and permutation tests are the more conservative inferential objects. The training-window sweep (Figure 11) demonstrates a consistent monotonic pattern across five T values, providing additional corroboration.

Economic validation. The gap-revision regression uses the predictive state $\alpha_{t|t-1}$ (which does not incorporate the current observation), providing a genuine ex-ante gap measure. The result is cross-sectional (it flips sign under actor fixed effects), not within-actor predictive.

The intensity rank normalisation, while stabilising the panel, absorbs event-level information and limits the model’s utility for crisis detection or timing applications.

7 Conclusion

We present a spectral state-space framework for forecasting cross-sectional investment intensity. On a 146-firm US CapEx/Revenue panel, a 2-mode DMD basis with rolling recomputation achieves predictive $R^2 = 0.711$ under nested CV (8 windows; $\Delta R^2 = +0.042$ vs AR(1); bootstrap CI $[+0.031, +0.055]$; exact permutation $p = 0.004$) and $R^2 = 0.845$ on 2 frozen holdout windows ($\Delta R^2 = +0.026$, both positive). A PCA baseline using the same pipeline but replacing DMD with eigendecomposition of the sample covariance achieves $R^2 = 0.702$ —statistically indistinguishable from AR(1)—suggesting the advantage is consistent with DMD’s temporal snapshot formulation, though the comparison cannot fully separate lagged-dynamics capture from implicit SVD regularisation. The result rests on three principles:

Dual regularisation: both the observation covariance ($R = cI$, replacing N^2 parameters) and the transition matrix ($F = 0.99I$, replacing K^2 parameters) must be aggressively regularised. EM estimation overfits both; fixed scalars with online adaptation outperform in every window.

Temporal basis tracking: on the 93-actor structural panel ($K = 8$), the spectral basis rotates at 25.8° per quarter ($\sigma = 6.5^\circ$) with stable dimensionality. This rotation, missed by a static basis, accounts for the single largest modal R^2 gain ($+14.7\text{pp}$, CI $[+11.1, +17.2]\text{pp}$). Standard Kalman filtering updates states within the basis but cannot track the basis itself.

Two complementary panels: the structural panel ($K = 8$, 93 actors, CapEx/Assets) identifies the richer rotating modal geometry and the role of dual regularisation; the predictive panel ($K = 2$, 146 firms, CapEx/Revenue) shows that a low-dimensional rolling variant forecasts well. On the predictive panel, EWM demeaning provides the largest single gain ($+3.5\text{pp}$), with rolling basis update adding a further $+3.4\text{pp}$; the Kalman filter’s incremental contribution at $K = 2$ is marginal (Section 6.2).

Predictive model-implied gap estimates (using $\alpha_{t|t-1}$) are associated with subsequent cross-sectional CapEx-rank revision ($t \approx -6.4$ to -7.1 with clustered SEs), though this is a between-actor effect that reverses under actor fixed effects (Table 10). The transition dynamics use stable default scalars ($F = 0.99I$, $Q = 0.5I$) that performed consistently across all evaluation windows in this sample—only the spectral basis requires periodic re-estimation.

Cross-sectional pooling. In a fixed-configuration sweep, the framework’s advantage increases monotonically as the training window shortens, from near-zero at $T = 7$ years to $\Delta R^2 = +0.057$ at $T = 2$ years (10/10 windows; Figure 11). This pattern is consistent with the spectral basis exploiting $N \times T$ cross-sectional observations collectively, while per-actor AR(1) is limited to each actor’s own history.

Limitations. Beyond the PCA baselines in Table 3, richer pooled benchmarks (panel AR with firm fixed effects, reduced-rank VARs) have not been tested. The economic validation uses predictive gaps ($\alpha_{t|t-1}$), which is clean ex-ante, but the result is cross-sectional (flips under actor FE), not within-actor predictive. The holdout advantage ($\Delta R^2 = +0.026$) is smaller than the CV mean ($+0.042$), though with only 2 holdout windows this is uninformative.

Scope conditions. The headline predictive result is on a 146-firm US CapEx/Revenue panel. The framework also beats AR(1) on Revenue/Assets (259 firms, $+1.0\text{pp}$) and a multi-ratio panel (270 firms, $+2.8\text{pp}$), confirming it is not specific to a single intensity construction (Table 8).

The structural findings (ablation, rotation, economic validation) are established on a broader 93-actor CapEx/Assets panel using modal (filtered) reconstruction. Return-based intensity (UK equities) and asset-growth intensity (banks) produce weak or negative R^2 —the framework requires sufficient cross-sectional dynamics (low persistence). The PCA cross-sectional baseline is consistent with a DMD-specific advantage, but richer pooled benchmarks (panel regressions with firm fixed effects, reduced-rank VARs) are a natural direction for future work. Generalisation to international markets and sub-quarterly frequencies remains open. The methodological lesson—dual regularisation and basis tracking for high-dimensional state-space models—extends beyond this specific application.

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Disclaimer. This paper presents a research framework for academic investigation of investment dynamics. It is not investment advice. The models, gap estimates, and statistical findings described herein have not been validated for live trading or portfolio construction. Past out-of-sample performance on historical data does not guarantee future results. The author bears no responsibility for any financial losses incurred by parties who deploy this framework or its derivatives in investment strategies. Any such deployment is undertaken entirely at the user’s own risk and should be subject to independent due diligence, regulatory compliance review, and appropriate risk management.

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